# GRAPHIC COMPUTING AND INTERACTION HUMAN-COMPUTER

## PROJECT
## XOCHIMILCO ECOLOGICAL PARK


Group 2

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# INTRODUCTION

This project aims at the three-dimensional reconstruction of the Xochimilco Ecological Park, integrating all stages of the graphics pipeline for its implementation in OpenGL. Based on real references from the site, representative elements were designed such as chinampas, trajineras, green areas, bodies of water, urban structures and furniture.

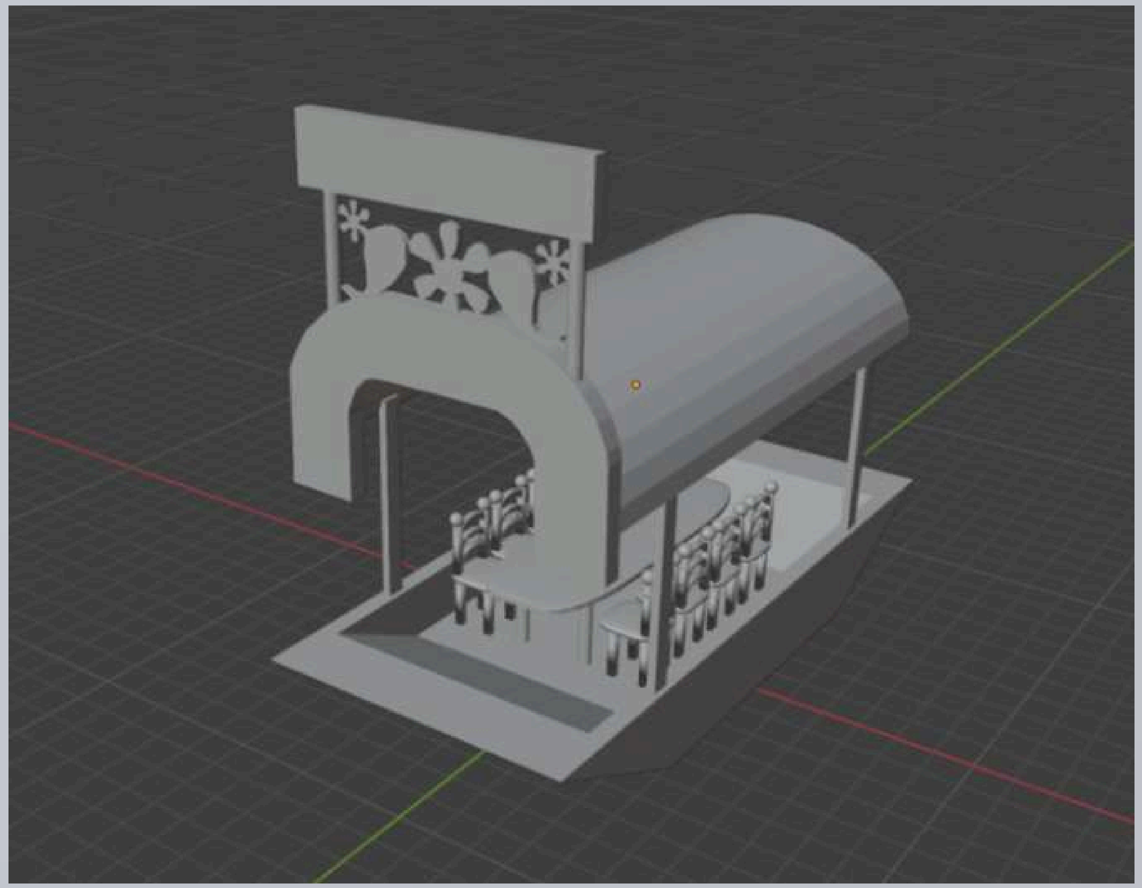
The development considered key aspects such as modeling, texturing, lighting, materials and camera, focused on achieving a coherent, visually appealing scene and optimized for real-time execution. The final scene was exported to an interactive environment, allowing an immersive experience that highlights the ecological and cultural value of the park.

# MODELING

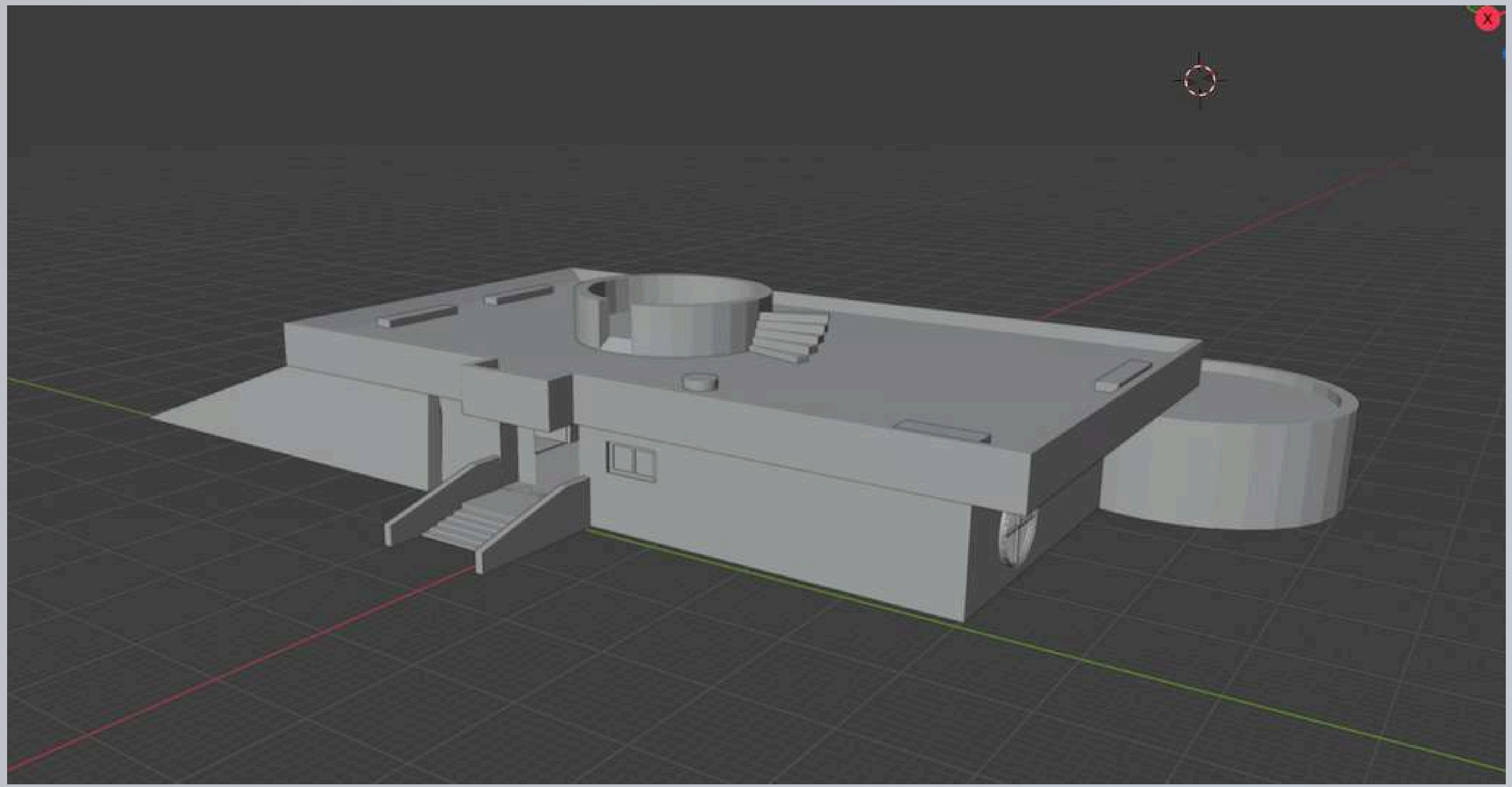
The representative elements of the park were modeled using Blender, starting from geometric primitives and subdivision, extrusion and digital sculpting. Among the highlighted objects are the following:

Park sign

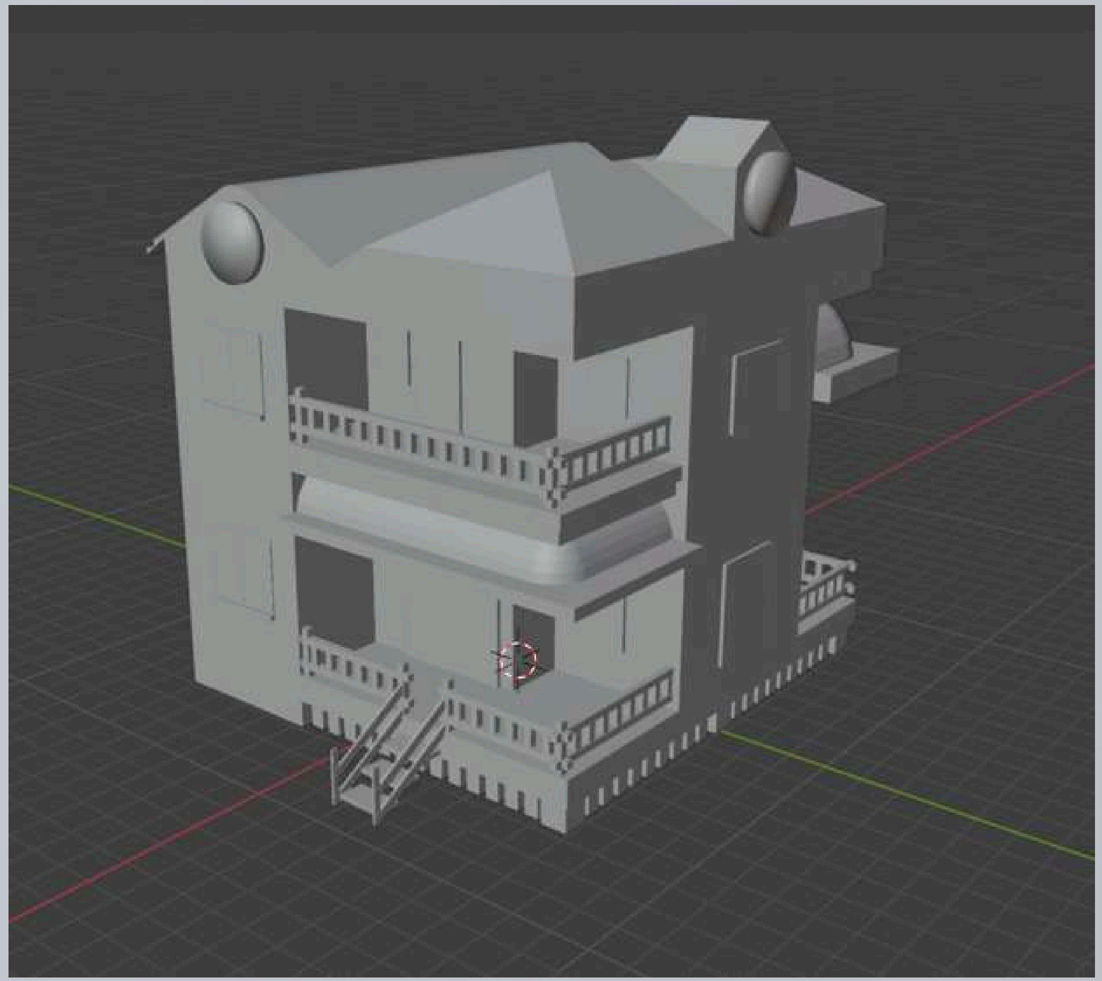

Retiro boat



Museum

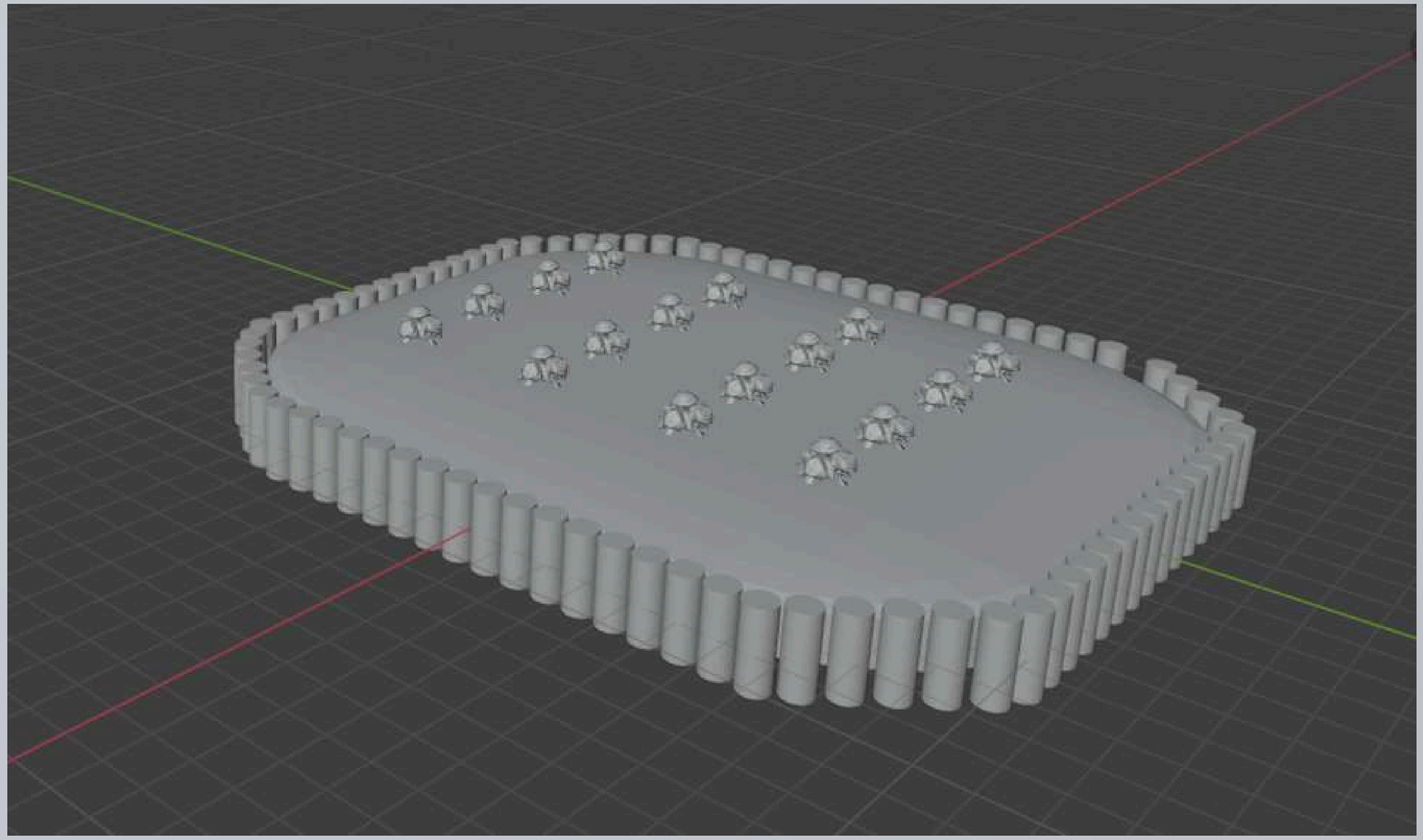


house



## Birds

## Chinampas

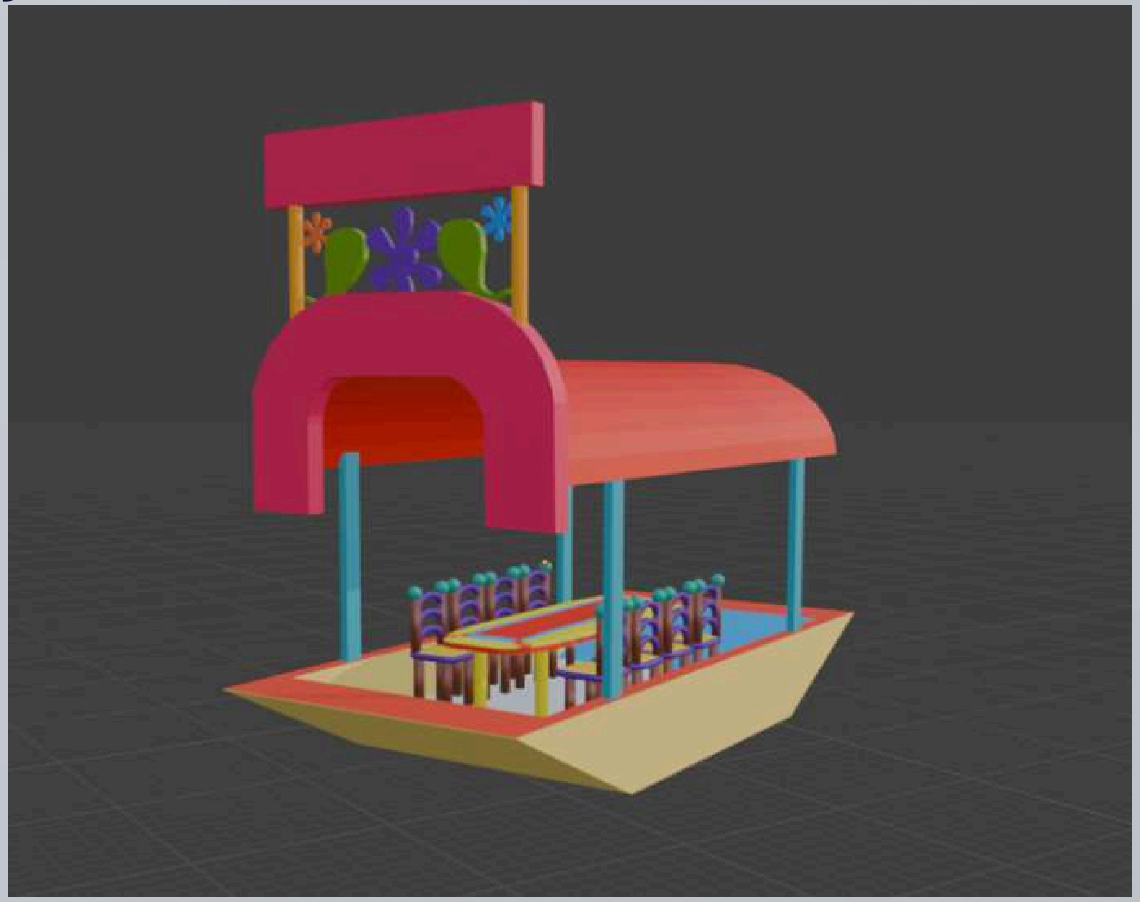


# TEXTURIZED

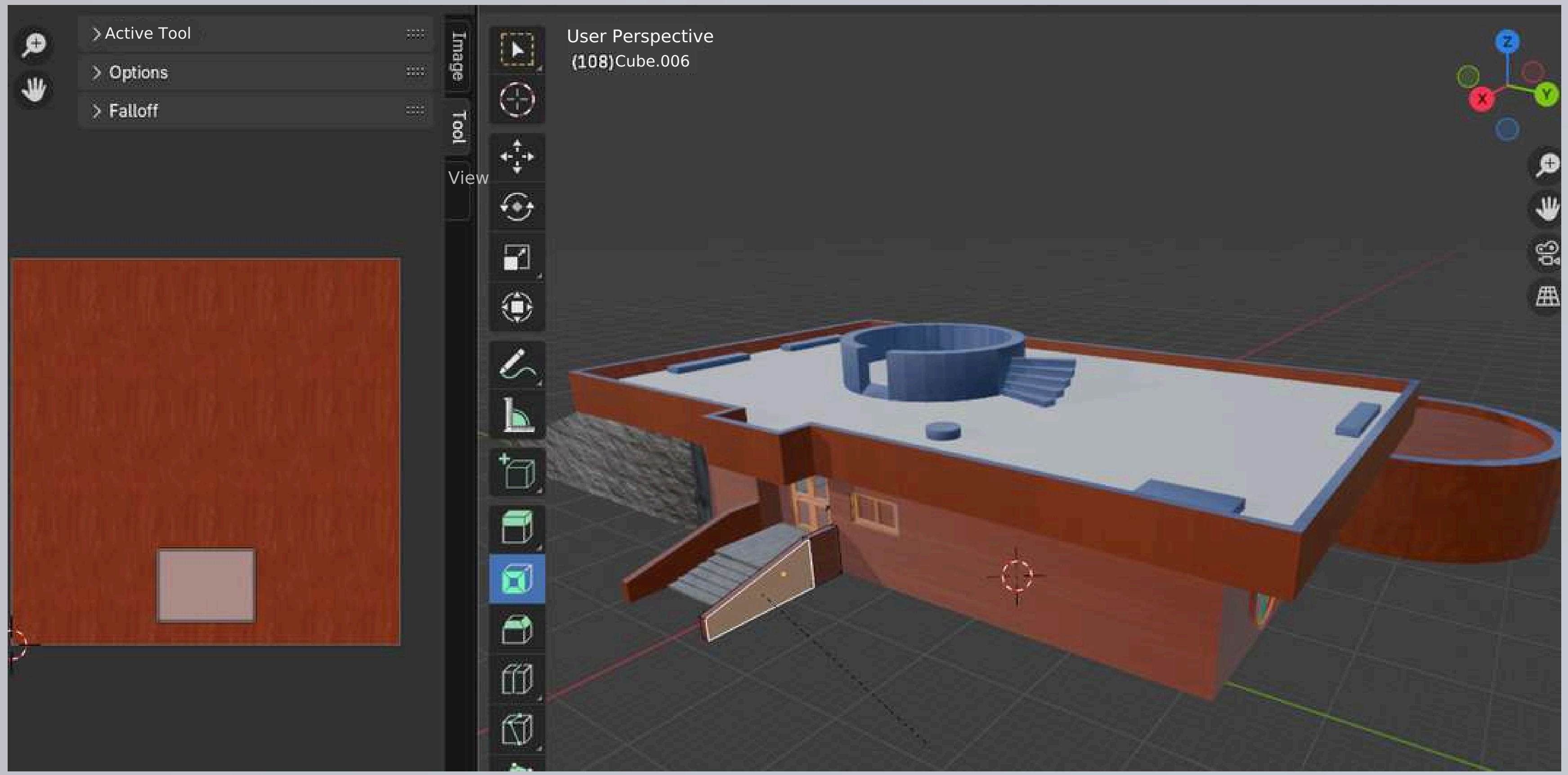
For texturing, specific images were selected that would represent the surfaces of the models, such as stone, earth, or wood. Thanks to the use of UV mapping, it was possible to apply these textures directly onto each model. After unwrap, the image arrangement on each face of the object was manually adjusted, ensuring it was well positioned, with no distortion or pixelation.

Trees


Trajinera



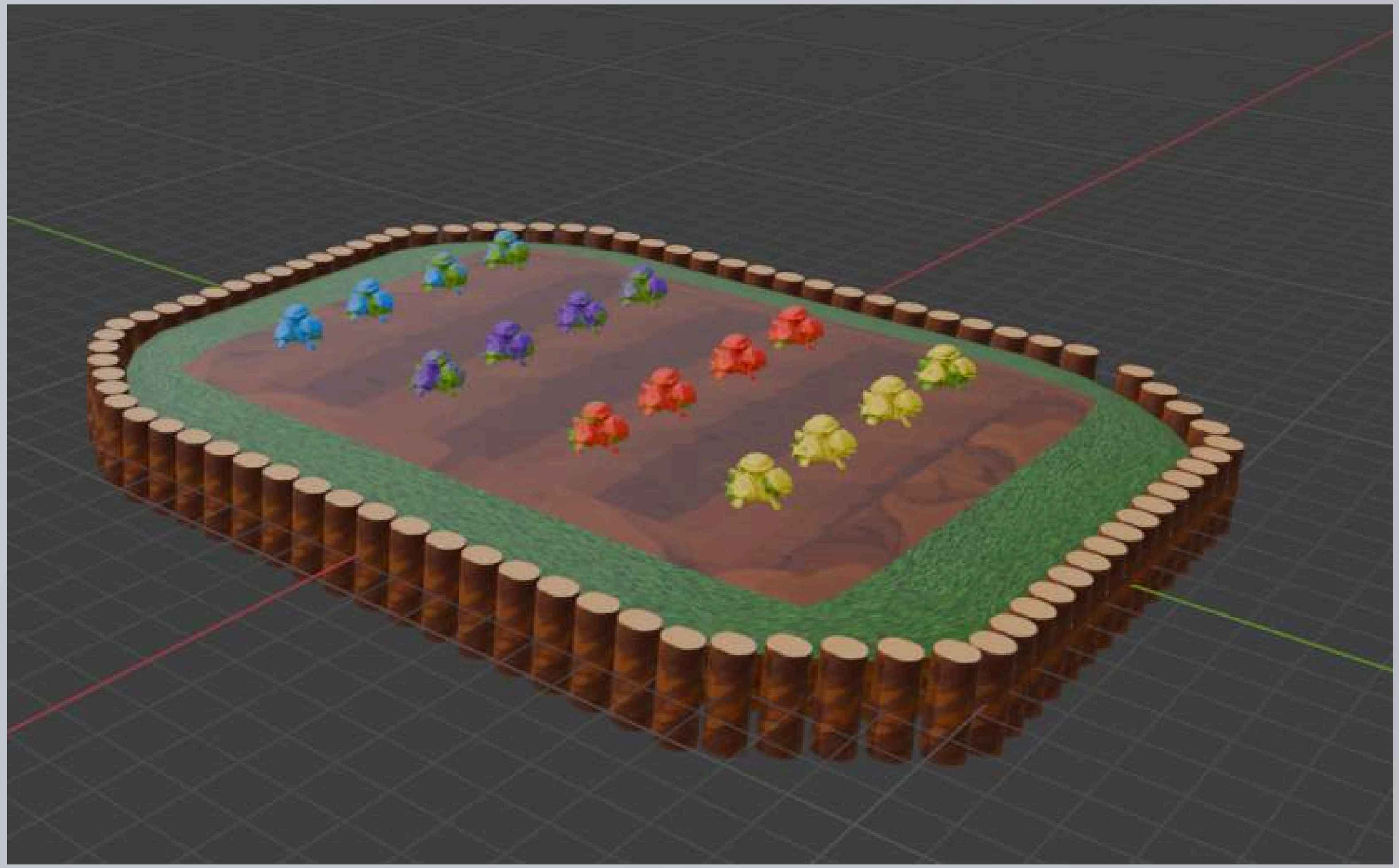
# Museum




# house

## Birds

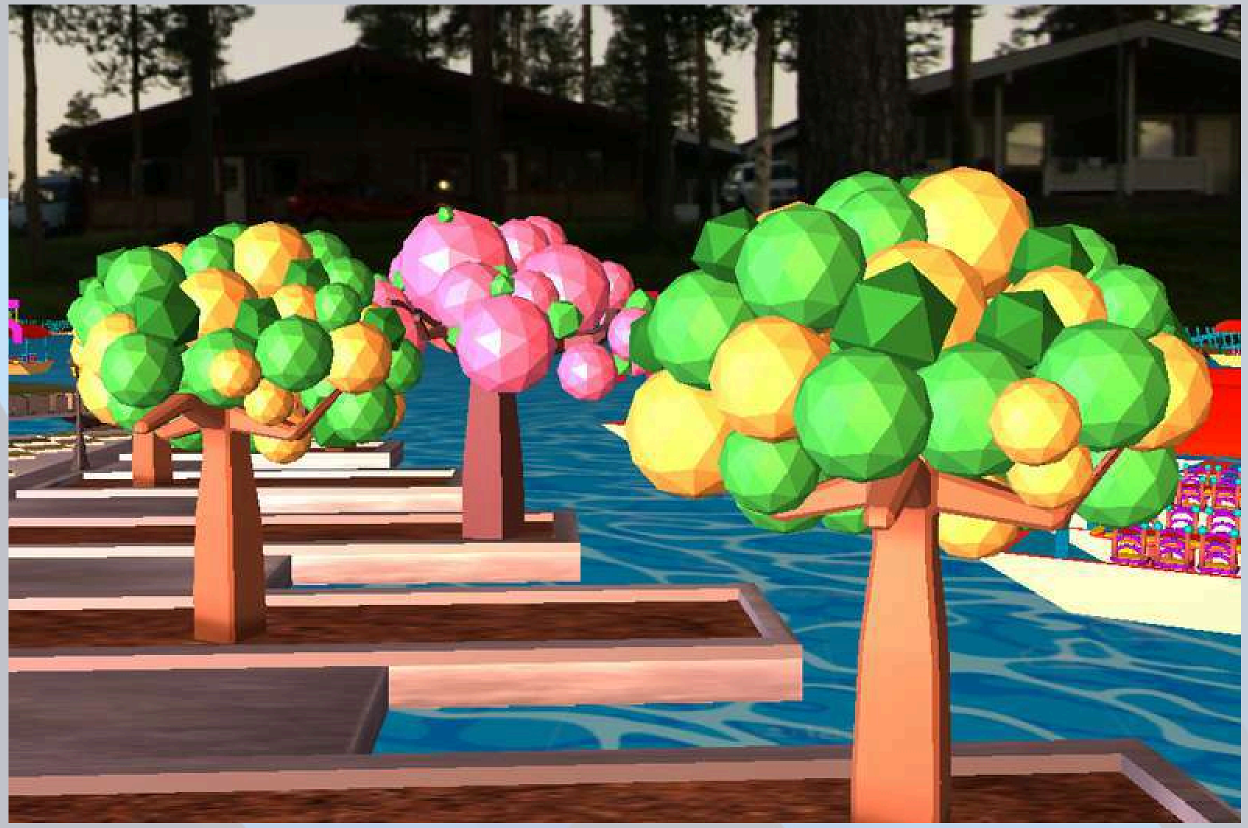
## Chinampas



# ILLUMINATION

The scene lighting was implemented directly in OpenGLA through shaders, the way light affects each surface was controlled, generating shadows, highlights and contrasts that provide visual depth. This system allowed lighting the park coherently and dynamically, highlighting materials, textures and volumes in real time during rendering.

Trees



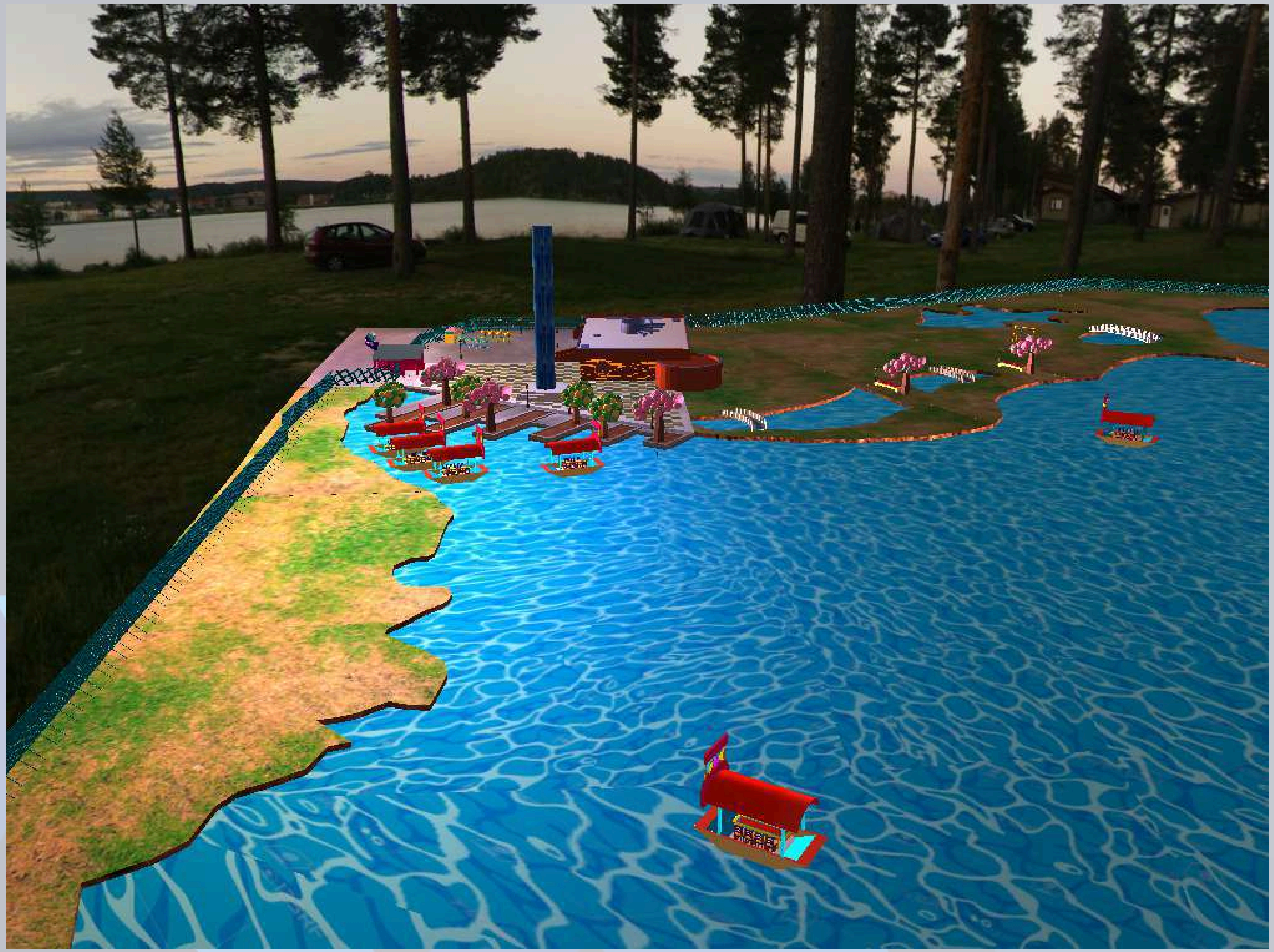
Bridge

# LIGHTING

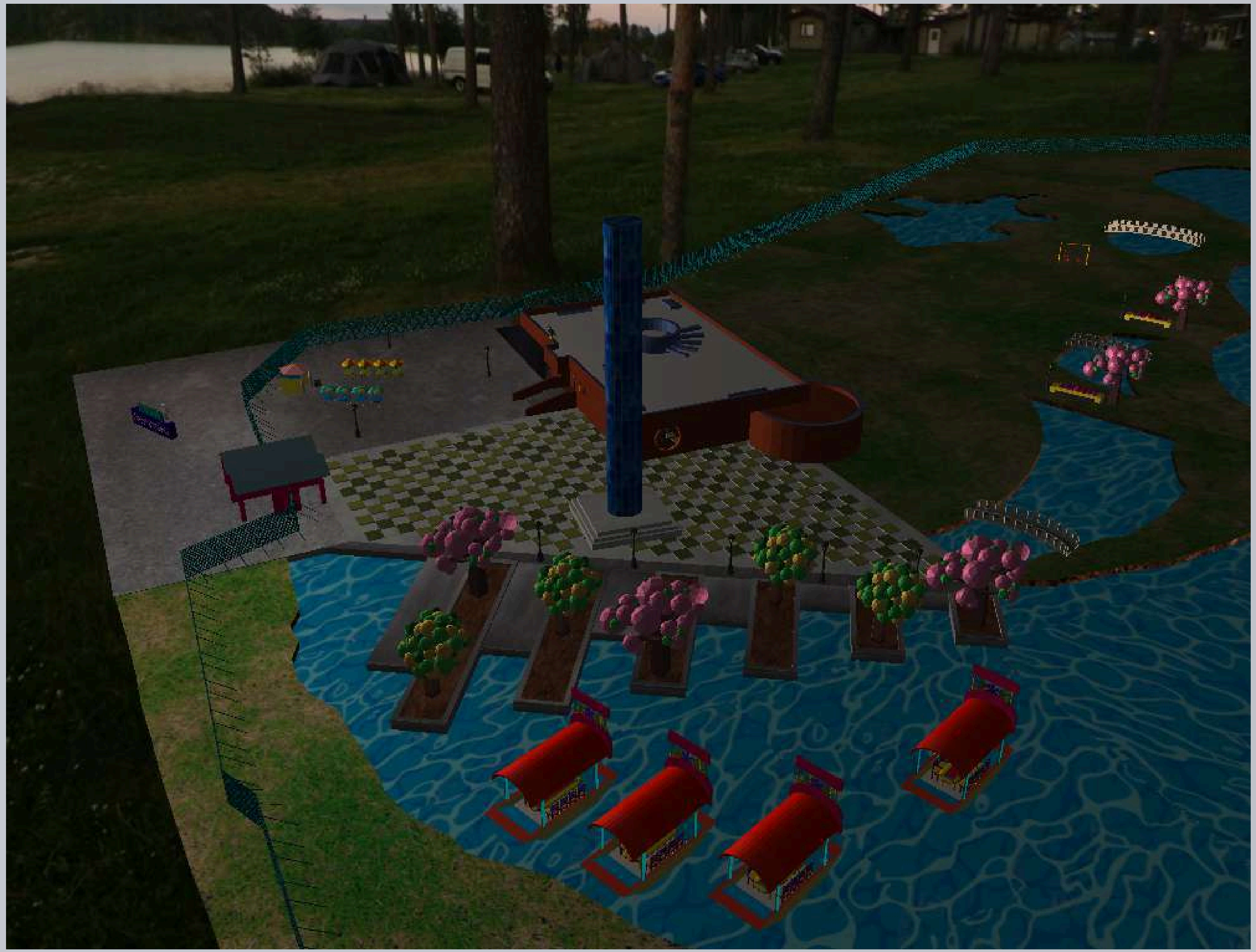
Museum interior

Chinampas area

With lighting



Without lighting

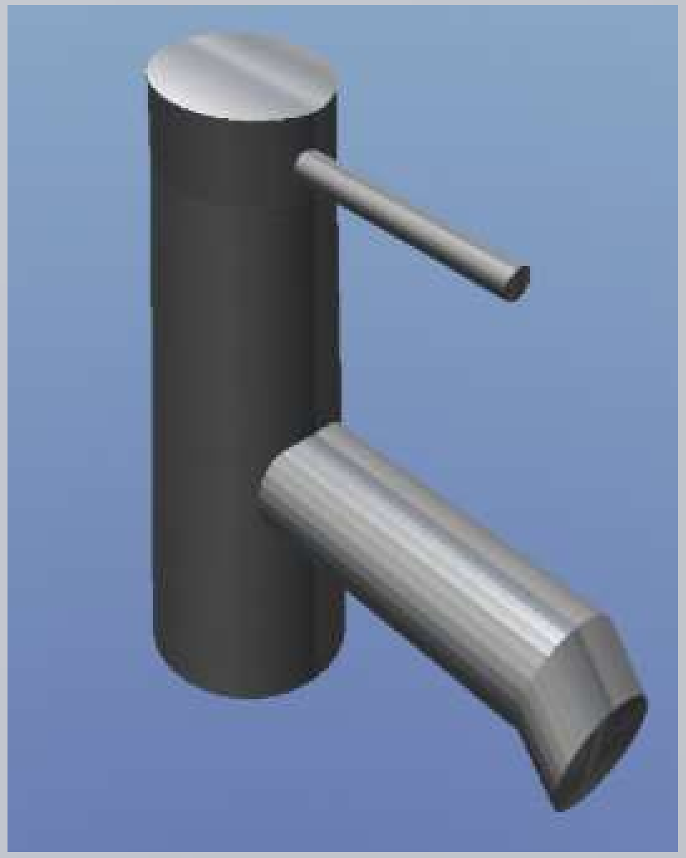


# MATERIALS

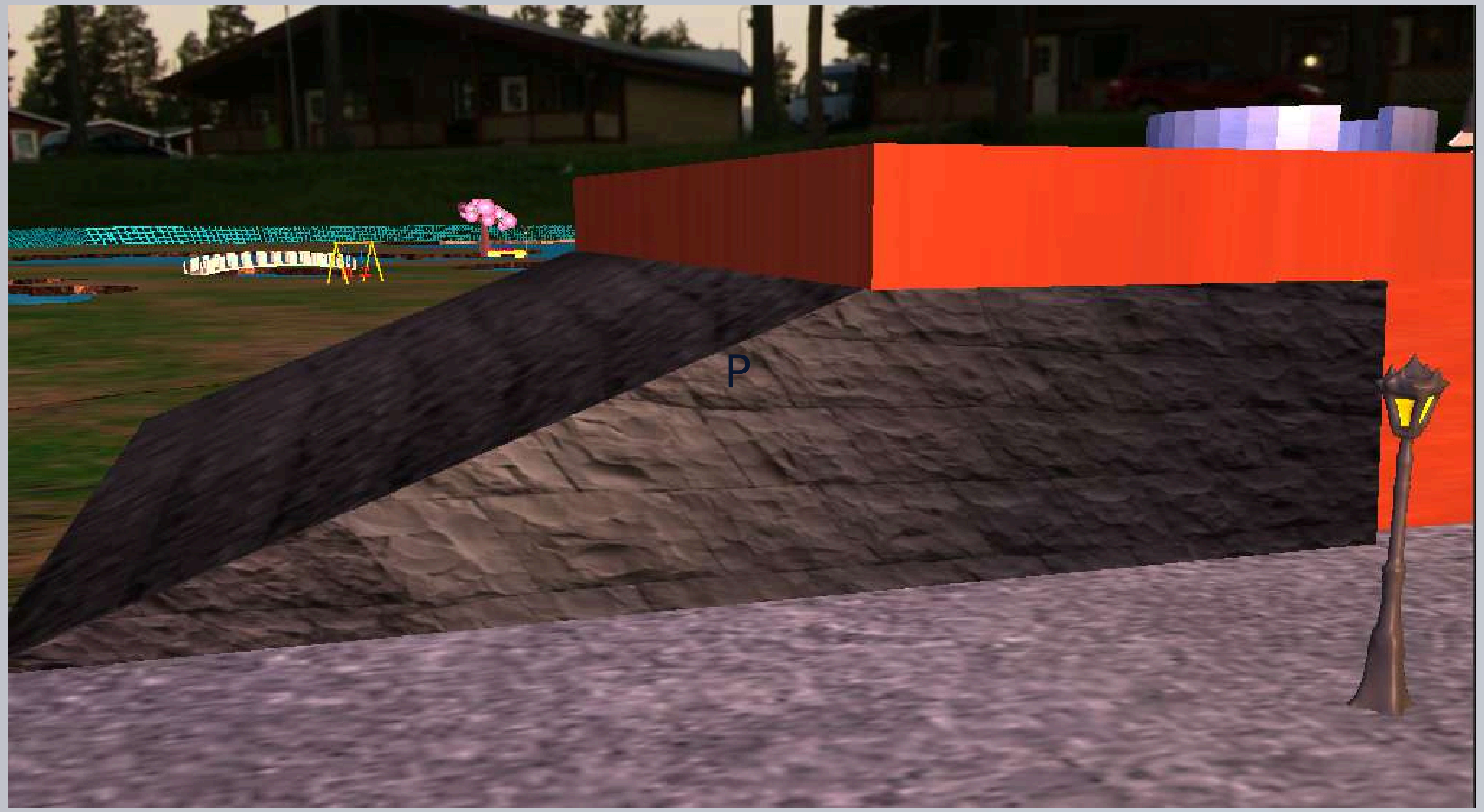
In the representation of materials, two approaches were combined. For some objects textures were used that directly simulated the desired material, such as wood, water or stone, so that visually it would be immediately recognizable. In other cases effects were applied through code in OpenGL to modify visual properties such as gloss, transparency or reflection, achieving that the object would react to light as the real material would.

Wooden bench

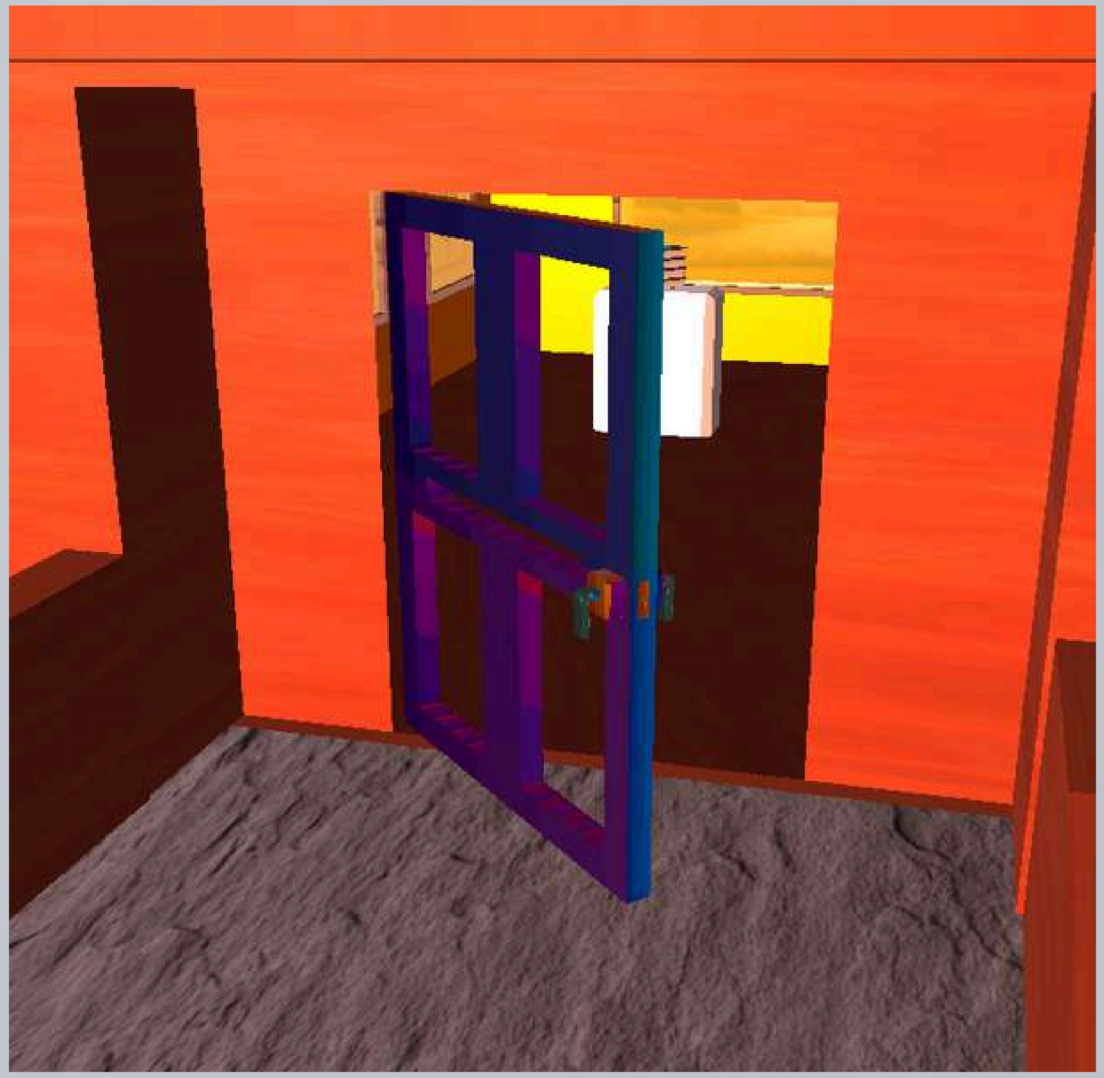
Metallic sink



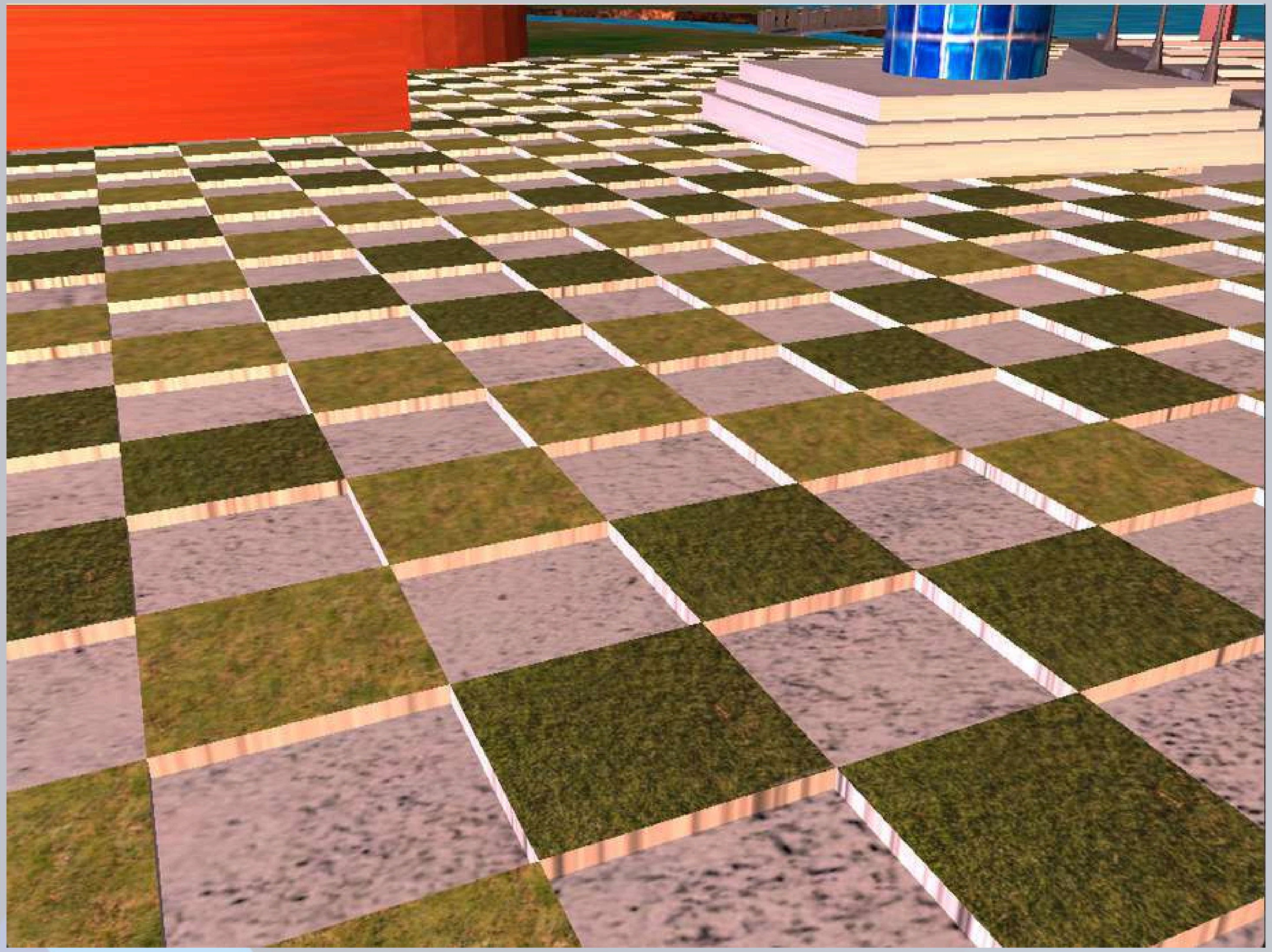
## Stone pyramid



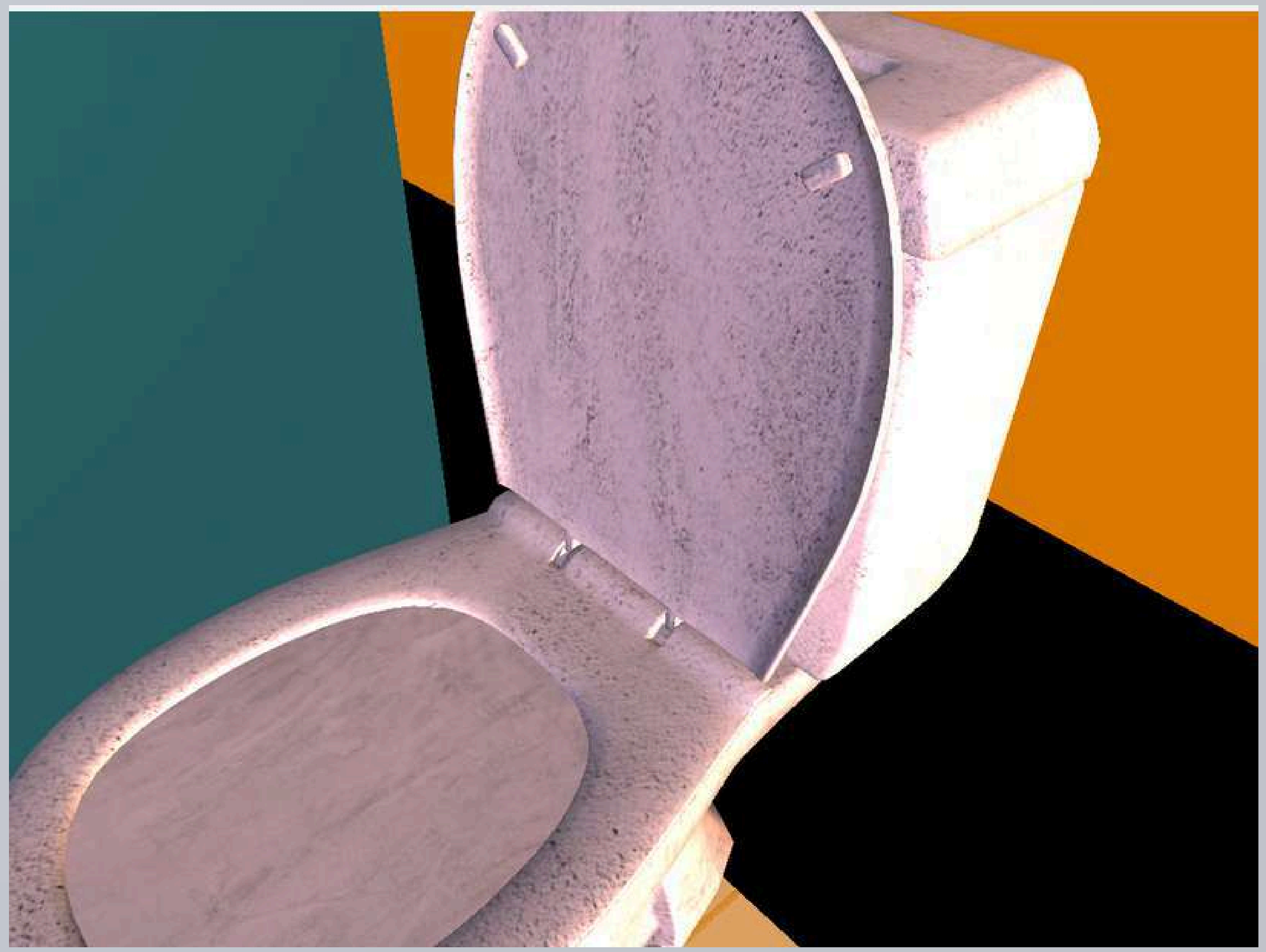

## Metal door



Hail floor



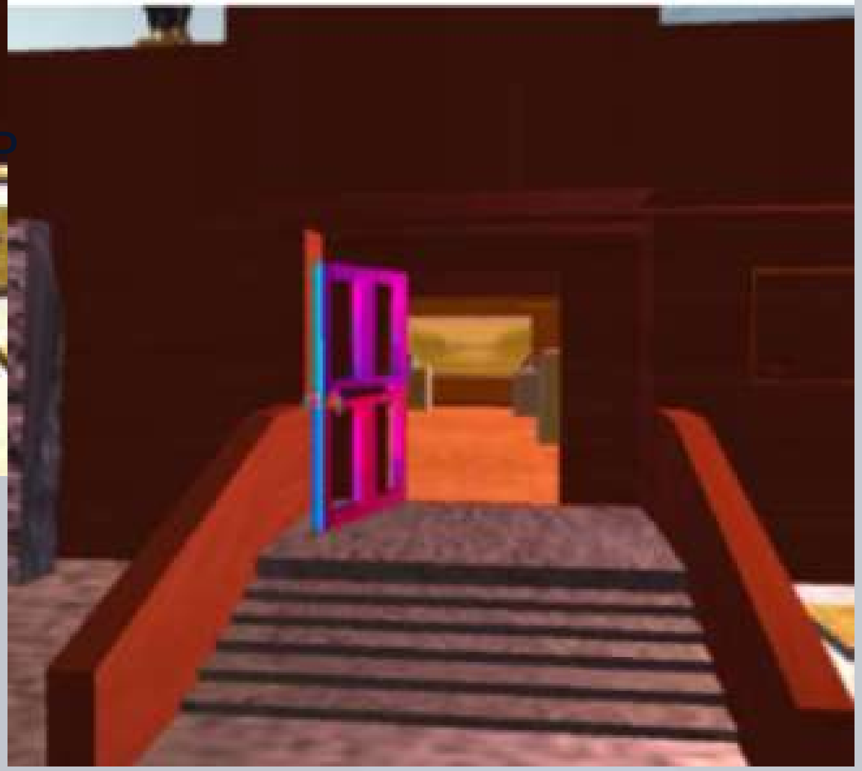
WC



# ANIMATION

For the animation part, the keyframe animation technique was implemented when exporting a model, created by us, to the Make Human platform. Likewise, we animated the plane that would serve as the water of the lake in our scene using the functions studied in class; in the same way, other animations were implemented, obtaining the following results:

## Main museum entrance



## Fan movement

Boatman? Or taxi boat? In Mexican canales trajinera refers to colorfully painted boats used in Xochimilco. Translation:


## Ghost appearance

## Bathroom door

## Lake movement

# FINAL RESULT

# CONCLUSIONS

- We understood how to organize a step-by-step computational graphics project.
- The stage-by-stage methodology allowed us to progress clearly and orderly.
- We learned to model, texture, animate, and program with greater fluency.
- We divided tasks according to each member's abilities.
- OpenGL was essential to building a coherent visual experience.
- We faced technical challenges that we solved by working as a team.
- We were able to represent an environment that promotes environmental awareness.
- We investigated the real ecological problems of Xochimilco.
- We developed a tour that informs, but also sensibilizes.

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